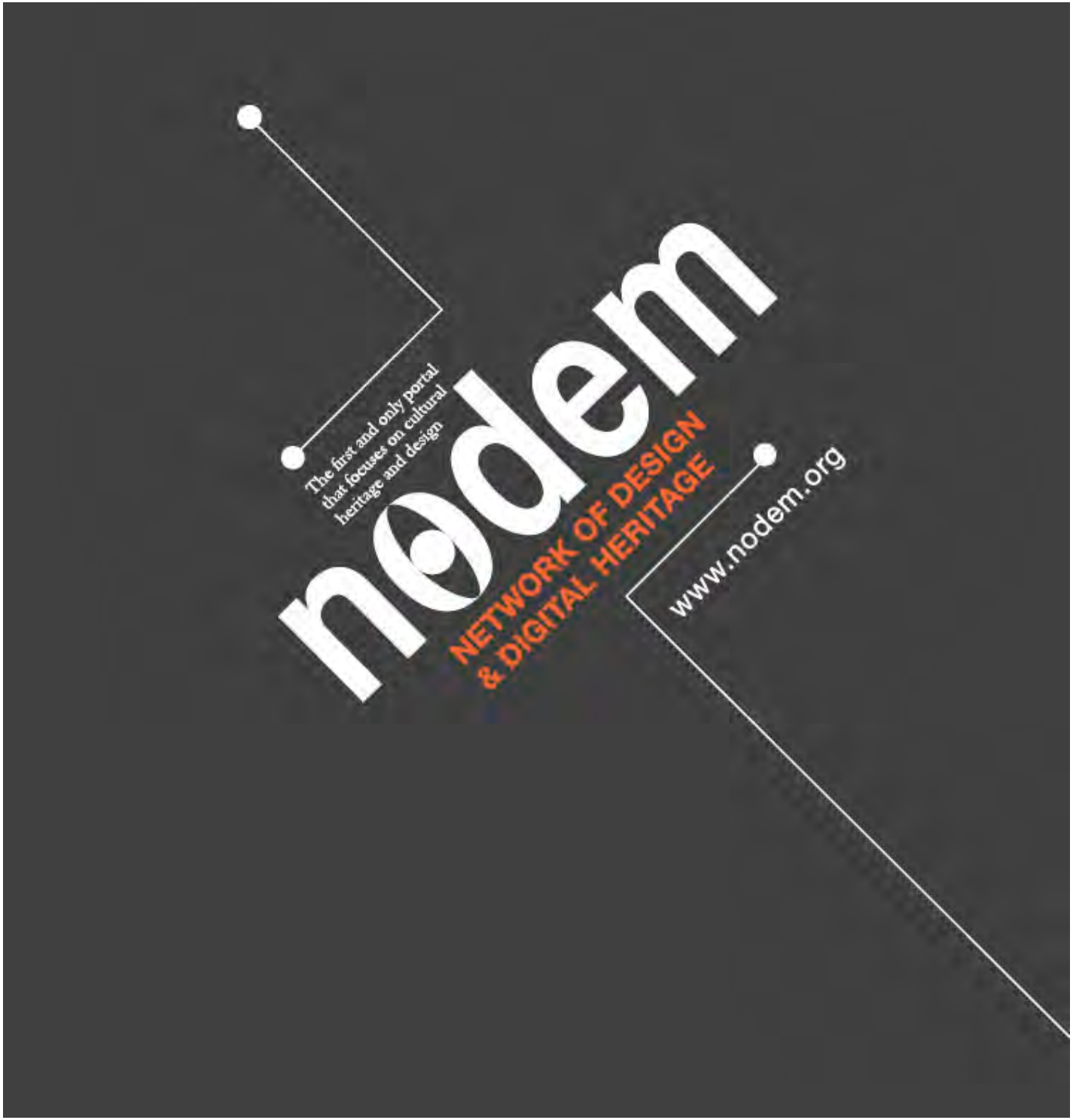
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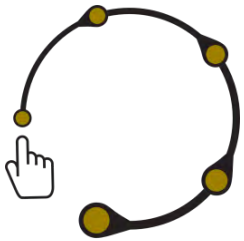
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# Virtual Museums in the 21st Century: Present Challenges and Future Prospects

**Susan Hazan** and **Sorin Hermon**



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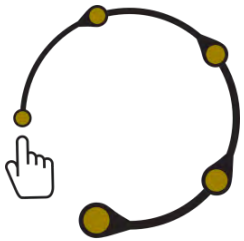
Virtual **MUS**eum **T**ransnational **NET**work

A network dedicated to Virtual Museums

Network of Excellence financed by the FP7

4 years: February 2011 – February 2015
18 Partners from 13 countries

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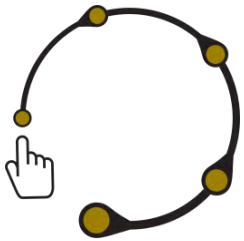
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## Partners

- CNR - Consiglio Nazionale delle Ricerche (ITABC – ISTI – ITD), Italy
- APRE: Agenzia Per La Promozione Della Ricerca Europea, Italy
- King's College London, Centre for Computing in the Humanities, King's Visualisation Lab, UK
- University of Sarajevo, Dept. Computer Science, Bosnia-Herzegovina
- Institut National de Recherche en Informatique et Automatique (INRIA-IPARLA Joint Research Team)
- Lund University, Department of Design Sciences, Sweden
- STARC, Cyprus Institute, Cyprus
- CINECA, Italy
- Foundation of the Hellenic World, Greece
- Allard Pirson Museum, University of Amsterdam, Netherlands
- Center for Documentation of Cultural and Natural Heritage< Egypt
- Comune di Roma, Sovraintendenza ai Beni Culturali, Museo dei Fori Imperiali, Italy
- Fraunhofer Institute für Graphische Datenverarbeitung, Germany
- Virtualware< Spain
- Visual Dimension, Belgium
- Sociedad Española de Arqueología Virtual, Spain
- Noho LTD, Ireland
- University of Brighton, University of Brighton's Business School, UK

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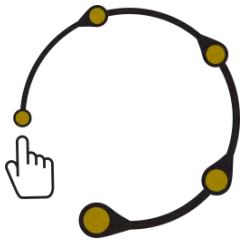
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## Associate Partners – Museums and Visitor Centers

Cyprus Department of Antiquities
Muzej Hercegovine
Muzej Sarajeva
Zemaljski muzej Bosne i Hercegovine
Nubian Museum, Aswan
Waterford Museum of Treasures, Dublinia and the Viking World
Museum Victoria, Melbourne
Historiska Museet Lund
Kulturen (Kulturhistoriska foreningen for sodra Sverige)
Bologna Archaeological Museum
Museo Civico del Risorgimento/Certosa, Bologna
Music Museum, Bologna
Polo Museale Fiorentino, Galleria dell' Accademia
Provincia di Napoli, Direzione Cultura
Roman National Museum,
Terme di Diocleziano museum
Teramo Archaeological Museum
Exposición Galiciadixital
Ayuntamiento de la Rinconada
Museo de Artes Tradiciones Populares, Madrid
Fundación Atapuerca
Fundación Marq-Museo Arqueológico Provincial de Alicante
Museo y Parque Arqueológico Cueva Pintada

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## Associate Partners

### Research Institutions

Applied Laboratory of Interactive Visualization and Embodiment, Hong Kong
CNR ISTC, Rome
CNR IBAM, Lecce
TU Delft (Technical University Delft)
University of Padova, Dept. Archaeology
University of California, Merced

### Companies

CadLand s.r.l.
MakeBelieve design & consulting
Playable Fiction
Zero Creative

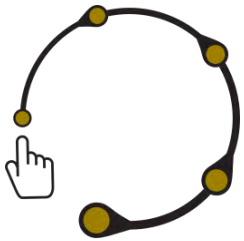
### Industry

HP Italy
NVIDIA South Europe

### International Institutions

UNESCO
ICOM

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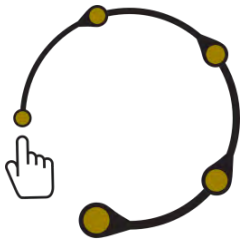
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## Goals

V-MusT.net will take into consideration and integrate results achieved by previous and on-going networks and IP projects related to the domain

- EPOCH
- STELLAR
- EUROPEANA LOCAL
- 3D-COFORM
- V-City
- Athena
- Carare
- GaLa
- STACHEM
- 3DTV
- 3D4YOU
- F-MUSEUM
- INTUITION

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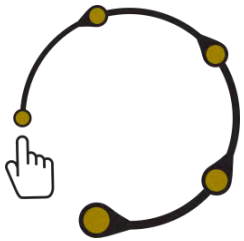
## Goals

V-Must brings together scientist from various domains to utilize their experience and knowledge for creating the virtual museum research network

The network is laying down the theoretical foundations of the virtual museum

V-Must identifies research areas for further development and is defining ideal scenarios for the Virtual Museum of the Future.

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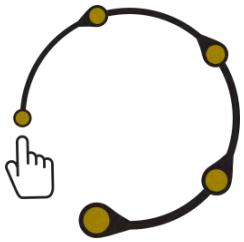
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## Data visualization and interaction

Research on data visualization and interaction techniques designed for:

- enhancing the users level of experience
- new forms of interaction with virtual objects
- reflecting cognitive and perceptive capabilities of users and enhancing the level of experience of the VH
- an integration of new mixed reality interfaces
- a wider dissemination of VH objects
- introducing pro-active user interfaces and its methodologies instead of symbolic/iconic systems

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## Questions

When is a virtual museum not a virtual museum ?

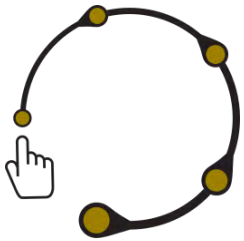
Does every museum needs to have its own dedicated virtual component ?

Does a virtual museum actually need a physical component?

Clearly the impressive *ALiVE* experiences have a robust life of their own; either with or without a physical anchoring in a permanent building.

Our research looks at these kinds of experiences, and by dint of their inclusion into the V-Must Network means identifying these kinds of experiences - by default - a virtual *museum*.

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## The Virtual Museum and Cultural Heritage

According to ICOM description on cultural heritage:

*Cultural heritage does not only embrace the tangible expressions like monuments and objects throughout the years. It also includes the living expressions like the traditions that many groups and communities worldwide have been passed down by their ancestors and will continue to pass on to their descendants, mostly by word of mouth. Although it is the motor of cultural diversity, this heritage is fragile.*

http://icom.museum/programmes/intangible-heritage

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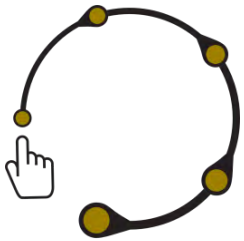
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## The Virtual Museum and Cultural Heritage

Clearly we are interested in the representation of cultural heritage, which can be at times extremely fragile as ICOM, the International Council of Museums identifies.

These are the kinds of questions that interest us in the V-Must project and our research leads us to pose several challenging questions.

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## The Core Concept

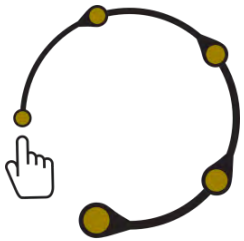
Does the virtual Museum by definition need to extend the mandate of the physical museum; as per the ICOM definition?

To recall the ICOM definition of a museum; education forms a central tenet of the *raison d'etre* of the museum.

According to the ICOM Statutes, adopted during the 21st General Conference in Vienna, Austria, in 2007:

> *A museum is a non-profit, permanent institution in the service of society and its development, open to the public, which acquires, conserves, researches, communicates and exhibits the tangible and intangible heritage of humanity and its environment for the purposes of education, study and enjoyment.*
>
> http://icom.museum/the-vision/museum-definition

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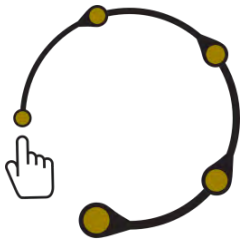
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Should then virtual museums be built just for education?

Would it be possible to develop a virtual museums for edutainment or even fun?

These are the kinds of questions that interest us in the V-Must project and we will address these issues in this presentation calling on a series of case studies that help us punctuate, and define our research.

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## Case Studies

These are the kinds of questions that interest us in the V-Must project and we will address these issues in this presentation calling on a series of case studies that help us punctuate, and define our research

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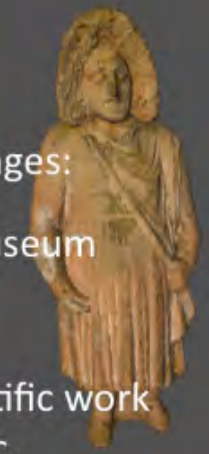
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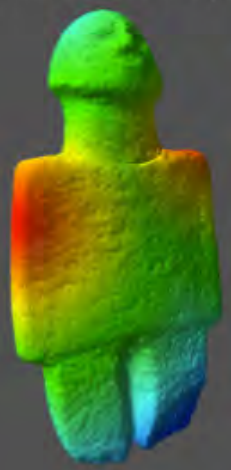




The The Cyprus Virtual Museum addresses the following challenges:

- Virtual re-contextualization of museum objects (not only museum exhibits, but objects once in use by people in the past)
- "Behind the scene" of exhibits – representation of the scientific work of scientific staff – conservators, curators, archaeologists, etc.







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The The Cyprus Virtual Museum implements the following technologies:

1. High definition and accurate 3D models
2. iPad based platform for investigating 3D models and related metadata.
3. Possibility to annotate 3D models
4. Personalized access to information (based on user types – advanced, researcher, etc.).
5. Augmented reality for bringing-up information.
6. Rich semantic layers of descriptive metadata
7. 3D semantic object-repository

